

Automated Kidney Stone Composition Analysis with Photon-Counting Detector CT, a Performance Study-A Phantom Study.

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For treatment of urolithiasis, the stone composition is of particular interest, as uric acid (UA) stones can be treated by chemolitholysis. In this ex vivo study, we employed an advanced composition analysis approach for urolithiasis utilizing spectral data obtained from a photon-counting detector CT (PCDCT) to differentiate UA and non-UA stones. Our primary objective was to assess the accuracy of this analysis method. A total of 148 urinary stones with a known composition that was measured by the standard reference method infrared spectroscopy (reference) were placed in an abdomen phantom and scanned in the PCDCT. Our objectives were to assess the stone detection rates of PCDCT and the accuracy of the prediction of the stone composition in UA vs non-UA compared to the reference. Automated detection recognized 86.5% of all stones, with best detection rate for stones larger > 5 mm in diameter (95.4%, 88.8% for stones larger than 3 mm, 94.7% for stones larger than 4 mm). Depending on the volume, we found a recognition rate of 92.8% for stones larger than 20 mm³ and 94.0% for stones with more than 30 mm³. Prediction of UA composition showed an overall sensitivity and a positive predictive value of 66.7% and a specificity and negative predictive value of 94.5%. Best diagnostic values volume wise were found by only including stones with a larger volume than 30 mm³, there we found a sensitivity of 91.7%, and a specificity of 92.4%. Sensitivity in dependance of the largest diameter was best for stones larger than 5 mm (85.7%), but specificity decreased with increasing diameter (to 91.3%). Automated urinary stone composition analysis with PCDCT showed a good automated detection rate of 86.5% up to 95.4% depending on stone diameter. The differentiation between non-UA and UA stones is performed with an NPV of 94.5% and a PPV of 66.7%. The prediction probability of non-UA stones was very good. This means the automatic detection and differentiation algorithm can identify the patients which will not profit from chemolitholysis.

Correlation and Prediction of IL-10Ra and DKK-4 Plasma Levels in Patients with Calcium Oxalate Urolithiasis.

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Calcium oxalate urolithiasis, commonly referred to as kidney stones, is a prevalent condition marked by the development of solid crystals within the urinary system. Currently, there is no established treatment to cure calcium oxalate urolithiasis. The underlying processes of calcium oxalate urolithiasis remain unclear, and the aim of this study was to clarify the correlation and prediction of IL-10Ra and DKK-4 plasma levels in patients with calcium oxalate urolithiasis. In this study, we explored the role of body mass index, eating habits score, levels of IL-10Ra and DKK-4, C-reactive protein, white blood cell count, blood urea nitrogen, and blood uric acid in HCC. In total, 85 patients with COU, attending our hospitals between January 2022 and June 2023, were enrolled in this study as experimental group (n = 85), and 85 healthy individuals, who underwent physical examinations during the same period, were collected as control group (n = 85). The obtained blood samples were collected for further testing. Numerous assays, including ELISA assay, western blot, and qRT-PCR, were employed to investigate the role of IL-10Ra and DKK-4 in calcium oxalate urolithiasis. The results indicate that the upregulation of IL-10Ra and DKK-4 may accelerate the progression of calcium oxalate urolithiasis. The logistic regression analysis indicates that the levels of IL-10Ra and DKK-4 positively correlated with calcium oxalate urolithiasis. In summary, the expression levels of IL-10Ra and DKK-4 positively correlated with the progression of calcium oxalate

urolithiasis, suggesting that IL-10Ra and DKK-4 could serve as potential predictive factors and risk factors for calcium oxalate urolithiasis.

Therapeutic effects of probiotics and herbal medications on oxalate nephrolithiasis: a mini systematic review.

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The majority of all kidney stone cases are oxalate urolithiasis with a high risk of recurrence. Beside its widespread occurrence, kidney stones are characterized by severe complications and high treatment costs. Probiotics and herbal medications could be forthcoming therapeutic interventions in the management of oxalate kidney stones. The PubMed/MEDLINE database was searched for keywords "Oxalobacter formigenes" AND "Oxalate" OR "oxalate degradation" AND "Lactobacillus" OR "Bifidobacterium" OR "recombinant Lactobacillus" OR "Bacillus subtilis", and "urolithiasis" AND "herbal extract". The search returned 253 results, 38 of which were included in the review. Most of the oxalate-degrading probiotics belong to the Oxalobacter formigenes, Lactobacillus, Bifidobacterium, and Bacillus genus with a minimum dosage of 107 CFU in the form of capsules, sachets, and lyophilized powder. Oxalate concentration in media was 5-50mM with an incubation time ranging from 24h to 14 days. The majority of the studies suggested that probiotic supplementation might be useful for reducing urinary excretion of oxalate and urea and alleviation of stone formation. Different herbal extracts were used on murine models of nephrolithiasis (induced by 0.5-3% ethylene glycol) with reduction of renal inflammation and urinary parameters, and calcium oxalate crystals. Several strains of probiotics and herbal extracts confer protective effects against kidney stone/nephrolithiasis, indicating their promising nature for being considered as elements of preventive / adjuvant therapeutic strategies.

Kidney Stone Disease: An Update on Current Concepts.

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Kidney stone disease is a crystal concretion formed usually within the kidneys. It is an increasing urological disorder of human health, affecting about 12% of the world population. It has been associated with an increased risk of end-stage renal failure. The etiology of kidney stone is multifactorial. The most common type of kidney stone is calcium oxalate formed at Randall's plaque on the renal papillary surfaces. The mechanism of stone formation is a complex process which results from several physicochemical events including supersaturation, nucleation, growth, aggregation, and retention of urinary stone constituents within tubular cells. These steps are modulated by an imbalance between factors that promote or inhibit urinary crystallization. It is also noted that cellular injury promotes retention of particles on renal papillary surfaces. The exposure of renal epithelial cells to oxalate causes a signaling cascade which leads to apoptosis by p38 mitogen-activated protein kinase pathways. Currently, there is no satisfactory drug to cure and/or prevent kidney stone recurrences. Thus, further understanding of the pathophysiology of kidney stone formation is a research area to manage urolithiasis using new drugs. Therefore, this review has intended to provide a compiled up-to-date information on kidney stone etiology, pathogenesis, and prevention approaches.

Urolithiasis, Urinary Cancer, and Home Drinking Water Source in the United States Territory of Guam, 2006-2010.

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We reviewed patient records with a first-listed diagnosis of urolithiasis-also known as urinary tract or kidney stone disease, nephrolithiasis-upon discharge from Guam's sole civilian hospital during 2006 to 2010 and urinary cancer mortality records from the Guam Cancer Registry for 1970 to 2009 to determine the source of municipal water supplied to the patients' residence. The objective was to investigate a possible relationship between the sources of municipal water supplied to Guam villages and the incidence of urolithiasis and urinary cancer. We analyzed hospital discharge diagnoses of urolithiasis or renal calculi by calculating the incidence of first-mentioned discharge for urolithiasis or renal calculi and comparing rates across demographic or geographic categories while adjusting by age, sex, and ethnicity/race. We reviewed cancer registry records of urinary cancer deaths by patient residence. The annual incidence of hospitalization for urolithiasis was 5.22 per 10,000. Rates adjusted for sex or age exhibited almost no change. The rate of 9.83 per 10,000 among Chamorros was significantly higher ($p < 0.05$) than the rates among any other ethnic group or race. When villages were grouped by water source, rates of patients discharged with a first-listed diagnosis of urolithiasis, adjusted for ethnicity/race, were similar for villages using either well water (5.44 per 10,000) or mixed source water (5.39 per 10,000), and significantly greater than the rate for villages using exclusively reservoir water (1.35 per 10,000). No statistically significant differences were found between the water source or village of residence and urinary cancer mortality. Some Guam residents living in villages served completely or partly by deep well water high in calcium carbonate may be at increased risk for urolithiasis compared with residents living in villages served by surface waters. Although the risk appears to be highest in villagers of Chamorro ethnicity, residents should be aware of other contributing risk factors and steps to take to avoid developing this health problem.

Identification of Novel Human Serum Albumin (SA) Inhibitors from Scoparia Dulcis for Urolithiasis.

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Urolithiasis is the process of forming stones in the kidney, bladder, and/or urinary tract. It has been reported that kidney stones are the third most common disorder among urinary diseases. At present, surgical procedures and Extracorporeal Shock Wave Lithotripsy (ESWL) are commonly employed for the treatment of Urolithiasis. The major drawback of these procedures is the recurrence of stones. This study aimed to identify potential natural inhibitors against human Serum Albumin (SA) from the plant Scoparia Dulcis for Urolithiasis. As protein-ligand interactions play a key role in structure- based drug design, this study screened 26 compounds from Scoparia Dulcis and investigated their binding affinity against SA by using molecular docking. The three dimensional (3D) structure of SA was retrieved from Protein Data Bank (PDB) and docked with PubChem structures of 26 compounds using PyRX docking tool through Autodock Vina. Moreover, a 3D similarity search on the PubChem database was performed to find the analogs of best scored compound and docking studies were performed. Drug-likeness studies were made using Swiss ADME and Lipinski's rule of five was performed for the compounds to evaluate their anti-urolithiatic activity. The results showed that citrussin c (Eugenyl beta-D-glucopyranoside) exhibited best binding energy of -8.1 kcal/mol with SA followed by aphidicolin, apigenin, luteolin and scutellarein. Two compounds (PubChem CID 46186820, PubChem CID 21579141) analogous to citrussin c were

selected based on the lowest binding energy. This study, therefore, reveals that these compounds could be promising candidates for further evaluation for Urolithiasis prevention or management.

Prospective study investigating the influence of nutritional intervention on biochemical profiles in patients with recurrent urolithiasis.

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Urolithiasis, commonly known as kidney stones, is a condition significantly impacted by dietary habits. The objective of this study is to evaluate the impact of a tailored dietary plan on the crystalluria and biological parameters of patients with different types of kidney stones over a 3-month period. We conducted a prospective study of 3 months. The study involved patients with recurrent nephrolithiasis. Alongside the medical consultation, a comprehensive dietary survey was performed to assess the patients' nutritional habits. Urinary parameters, including volume, calcium, oxalate, uric acid, and power of hydrogen (pH), were evaluated both before and after the dietary intervention. 69 patients were involved. There were 17 patients diagnosed with cystine lithiasis, 33 with oxalocalcic lithiasis and 19 with uric lithiasis. After 3 months, only 32 patients revisited for follow-up. There were significant changes ($p = 0.002$ and 0.04) in urine crystalluria for cystinic and uric lithiasis. For the urinary oxalate variation, there was a significant decrease from T1 (before dietary intervention) to T2 (after dietary intervention), with levels dropping from 0.289 ± 0.10 $\mu\text{mol/l}$ to 0.215 ± 0.079 $\mu\text{mol/l}$ ($p = 0.02$). Regarding urinary calcium (calciuria), there was a trend toward a decrease from T1 to T2, although the change was not statistically significant, with levels decreasing from 2.42 ± 1.68 $\mu\text{mol/l}$ to 2.14 ± 1.62 $\mu\text{mol/l}$ ($p = 0.1$). Our research underscores the favorable effects of a tailored and well-balanced diet on both the crystalluria and biological parameters of individuals with recurrent lithiasis.

Myricetin Attenuates Ethylene Glycol-Induced Nephrolithiasis in Rats via Mitigating Oxidative Stress and Inflammatory Markers.

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Urolithiasis or nephrolithiasis is a condition of kidney stone formation and is considered a painful disease of the urinary tract system. In this work, we planned to discover the therapeutic roles of myricetin on the ethylene glycol (EG)-induced nephrolithiasis in rats. The experimental rats were treated with 0.75% of EG through drinking water for 4 weeks to initiate the nephrolithiasis and subsequently treated with 25 and 50 mg/kg of myricetin. The body weight and urine volume were measured regularly. After the sacrifice of rats, the samples were collected, and serum and urinary biomarkers such as creatinine, urea, Ca^{2+} ion, and BUN, OPN, oxalate, and citrate levels were determined using assay kits. These biomarkers, the MDA level and CAT, SOD, and GPx activities, were assessed in the kidney tissue homogenates. The IL-6, IL-1 β , and TNF- α levels were also quantified using respective kits. The histopathological analysis was done on the kidney tissues. Myricetin treatment did not show major changes in the body weight and kidney weight in the EG-induced rats. The treatment with 25 and 50 mg/kg of myricetin considerably reduced the urea, creatinine, BUN, Ca^{2+} ion, and oxalate and increased the citrate content in serum and urine samples of EG-induced rats. Further, myricetin depleted the inflammatory cytokines and MDA levels and elevated the CAT, SOD, and GPx activities in the renal tissues. The activities of ALT, AST, ALP, GGT, and LDH were also reduced by the myricetin. Furthermore,

the myricetin upheld the histoarchitecture of the kidneys. The outcomes of this investigation propose that myricetin is effective in EG-induced urolithiasis probably because of its antioxidant, anti-inflammatory, and renoprotective activities. In addition, further studies are still required to verify the precise therapeutic mechanism of myricetin.

Urolithiasis. Terms and concepts.

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Urolithiasis is a general term that encompasses formulation of urinary tract stones as a sequela of one or more underlying abnormalities. A variety of terms have been used to describe the characteristics of uroliths based on mineral composition, location, and/or shape. Knowledge of the origins and meanings of these terms will facilitate understanding of the cause, detection, treatment, and prevention of uroliths.

Analysis of 451,891 canine uroliths, feline uroliths, and feline urethral plugs from 1981 to 2007: perspectives from the Minnesota Urolith Center.

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Urolithiasis is a general term referring to the causes and effects of stones anywhere in the urinary tract. Urolithiasis should not be viewed conceptually as a single disease with a single cause, but rather as a sequela of multiple interacting underlying abnormalities. Thus, the syndrome of urolithiasis may be defined as the occurrence of familial, congenital, or acquired pathophysiologic factors that, in combination, progressively increase the risk of precipitation of excretory metabolites in urine to form stones (ie, uroliths). The following epidemiologic discussion is based on quantitative analysis of 350,803 canine uroliths, 94,778 feline uroliths, and 6310 feline urethral plugs submitted to the Minnesota Urolith Center from 1981 to 2007.
